

Latest buzzers

Piezoelectric buzzers are widely used in applications ranging from consumer products (such as home appliances) to automotive and industrial equipment. The latest technological advancement in this regards is that, the buzzer is considerably getting smaller and lighter. The size and weight of the buzzer has significantly reduced up to 44 per cent. Sound-pressure-level output, measured 10cm from sounder, has become better. Maximum peak-to-peak voltage measured is $\pm 12.5V$. Power consumption is typically 0.6mW, which helps preserve battery life in power budget-constrained consumer electronic devices. These compact ultra-light and low-power piezoelectric sounder suits are gaining potential applications in healthcare devices like blood glucose meters and personal thermometers. Such a solution is being offered by companies like Murata Manufacturing.

The latest and upcoming trends in audio-video component shared in this article are expected to revolutionise primarily the consumer electronics industry, thus setting a benchmark by involving industry players to be part of this change.

FPGAs make their mark on the audio-video processing world

System designers have always been used to utilise discrete devices such as digital signal processors (DSPs) or graphics processing units (GPUs) to develop embedded vision systems. Programmable logic is now said to offer performance, cost and integration advantages at each step of the developmental process. Is this true?

Just a few days back, in mid-April, Blackmagic Design, a company that develops video equipment, announced that they were launching a new camera, which they claim to be the world's smallest ultra-HD 4K cinema production camera (at least, for now). Why did I mention this here? A Xilinx Kintex-7 field-programmable gate array (FPGA) forms the transceivers in its systems on chip (SoC). The SoC implements advanced video and image processing algorithms right within the FPGA. We are talking about a complete 4K camera SoC, which means horizontal resolution to the order of 4000 horizontal pixels.

What does this mean for designers? By using these programmable solutions to form the core of a video-processing solution, they are able to continuously refine algorithms that they use to process videos and images. Designers will also be able to keep iterating until they reach the targeted image and video quality.

Altera has its own video-design framework that provides a combination of programmable logic devices, IPs and tools to develop vision systems. These could be utilised to create low-cost high-performance video and image signal processing, integrating analytics into surveillance systems (motion detection, facial recognition, etc) or for implementing high-efficiency data compression.

HDMI2C2-14HD and HDMI2C1-6C1) designed for maximum performance.

Colour sensors. Conventional colour sensors required a transparent optical window in order to accurately detect visible (that is, RGB) light. However, the design trend in many smartphones and portable devices is to adopt dark optical windows for a smart outlook, making it difficult to detect colour. Since less visible light gets through, the light that permeates gets further mixed in with infrared (IR) rays. But the launch of new brightness colour sensor optimised for smartphones leverages the original IR-removal technology. IR cut-off characteristics like reducing the effects of IR rays by over 10x as compared with conventional prod-

ucts enables compatibility with dark (low-permeability) optical windows. Colour sensors for display-equipped devices such as smartphones and tablets are capable of detecting colour temperature, brightness and RGB components of ambient light. A product like ROHM's BH1745NUC sensor is one among the newer launches.

What is happening in audio

All-in-one chips. E-Call, instrument cluster, telematics and in-car listening experience are expected to get better with the launch of full-digital class D amplifiers. The full-digital amplifier architecture featured with built-in low-radiation function, floating digital core, anti-cross conduction and slope control minimises tuner

interference, enhances mobile phone immunity and simplifies compliance with all other emission standards, enabling high audio performance by using low-cost external components. These amplifiers connect noiselessly to the digital audio processor without a digital-to-analogue converter (DAC) or input filter. Companies like TI (TAS5421-Q1) and STMicroelectronics (FDA2100LV) are contributing their solutions in this sphere.

Better microphones. Often in loud and noisy environment it becomes difficult to hear the voice of the person on a mobile phone. Due to noise and other disturbances, communication is not proper. Now, with the launch of advanced MEMS microphones, one can hear better on mobile devices.

MEMS microphones maintain ultra-low distortion at less than 10 per cent up to very-high external sound-pressure levels, enabling equipment such as smartphones and wearable devices to perform better when placing calls or recording audio in loud environments. In MEMS technology, the preamplifier design is the key, as this prevents saturation of the output signal even when background-noise levels are high, such as at concerts, bars or clubs, or if the user is speaking loudly close to the microphone.

In addition, omni-directional sensitivity ensures excellent overall performance and versatility in mobile applications. STMicroelectronics MP23AB02B is one such high-quality microphone.

Enhanced headphones. In advanced headphones, a new chipset has been launched that draws low power and high-quality codec for portable and general-purpose audio applications. Traditionally, to 24-bit stereo ADCs and DACs, this new device integrates a broad range of additional functions to simplify implementation of complete audio system solutions. The device reduces external component requirement by integrating drivers for speakers, headphones and differential or stereo-line outputs, and integrates preamps for